How Many Documents Cover a Corpus

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Abstract

A sample drawn from a survey population comes with a margin of error and a formula for the sample size a target precision needs. A sample drawn from a corpus to annotate or to summarize comes with neither. We supply both for one question: what share of the corpus lies within a chosen cosine distance r of some sampled document, in a fixed embedding space. The share that does not is the missing mass at resolution r. Its estimator G needs no labels: it is the share of sampled documents with no other sampled document within r (Maurer, 2022). We show that G, as a function of sample size n and resolution r, depends on them only through nr^d, where d is the intrinsic dimension of the embedded corpus estimated from a pilot of 2,000 documents. The dimension d is therefore the corpus property that sets the sample size, in the role the variance plays in a survey formula. On three forum corpora and two social media platforms a fit on samples of at most 1,000 documents predicts G at 7,000 documents to a mean error of 0.02 to 0.05, and the sample size it prescribes for a target isolation rate lands within a factor of 1.5 of the empirical one in 20 of 24 tested cases. For a finite corpus of N documents the missing mass is $G\,(1 - n/N)$, the finite population correction of survey sampling, to a mean error of 0.01 to 0.02. On one corpus of political posts the dependence on nr^d fails because its clusters have local dimension from 2 to 16, so the fit's R^2 tells a user whether one number describes a corpus. A second question, how many documents are needed to see every theme, concerns a partition and is answered by the theme masses alone: for thirty themes the answer is the same curve in all eight corpora we tested. Ten real corpora from four sources, two encoders and three seeds support these results.

1 Introduction

Large collections of text are sampled before they are studied. An annotation team codes a few thousand posts out of millions, a language model summarizes a few hundred documents because the corpus does not fit its context, and a topic model is fitted to a slice because the full run is slow. The sampler of a human population has a settled answer to the question such a slice raises. Given the population variance σ^2 and a tolerated error e, the formula $n = z^2\sigma^2/e^2$ gives the sample size, and the estimate comes with a margin of error (Cochran, 1977; Kish, 1965). The sampler of a corpus has no such formula. The sample size is set by budget, and whether the slice covers the corpus is left unasked.

The question has an exact form once a corpus is a set of points. Embed each document, choose a distance r that says what counts as the same content, and ask what share of the corpus lies farther than r from every sampled document. We call that share the missing mass at resolution r. In the discrete case, where two documents either share a category or do not, it is the quantity Good and Turing estimated from the share of draws whose category was seen once (Good, 1953). Ecologists built a sampling theory on that estimate: a rarefaction curve gives the coverage a smaller sample would have had, an extrapolation curve gives the coverage a larger sample will have, and together they say how many more units a target coverage needs (Chao and Jost, 2012). Maurer (2022) carried the estimator into metric spaces. His statistic G is the share of sampled points with no other sampled point within r, and he proved bounds on its bias and variance. The estimator has had no application to text, and nothing in Maurer's treatment says how G falls as the sample grows.

This paper turns G into a sample size formula for corpora and tests the formula on ten real corpora. Four results carry the paper.

1. $G(n, r)$ depends on n and r only through nr^d, where d is the intrinsic dimension of the embedded corpus. A curve fitted on samples of

at most 1,000 documents predicts G at 7,000 documents to a mean error of 0.02 to 0.05 on five real corpora, and the fit recovers d on synthetic manifolds of known dimension. The sample size for a target isolation rate at resolution r then follows in closed form, and it lands within a factor of 1.5 of the empirical sample size in 20 of 24 tested cases on corpora that obey the law. The corpus property the formula needs is d, estimated from a pilot of 2,000 documents.

2. For a finite corpus of N documents the missing mass is $G\,(1 - n/N)$. The factor is the finite population correction of survey sampling, and it brings the mean error of G against the true missing mass from 0.07 to 0.015.

3. On one corpus of political posts the law fails. Its clusters have local dimension from 2 to 16, slogans beside discussion, and the failure persists after near duplicates are removed. The R^2 of the fit is therefore a diagnostic that says whether one dimension describes a corpus.

4. How many documents are needed to see every theme is a different question. Once themes are a partition of the corpus, the answer depends on the theme masses alone. With thirty themes per corpus it is the same curve in all eight corpora we tested, 0.82 to 0.86 of the theme mass at $n = 50$, while coverage at a fixed resolution runs from 0.003 to 0.77 across the same corpora at $n = 25$. The two questions need two tools, and we say which answers which.

In practice the procedure is short. A user picks the resolution r that matches the question, embeds a pilot, estimates d, and takes the sample size from the formula. After the sample is drawn, $G\,(1 - n/N)$ reports the share of the corpus it leaves uncovered. If the fit is poor the corpus has more than one scale, and the user is told so before the sample is drawn.

2 Background

2.1 Sample coverage in the discrete case

A sample of n draws from a population of categories covers the categories it has seen. The mass it has missed is the total probability of the unseen categories. Let f_k be the number of categories that appeared exactly k times in the sample. Good (1953) showed that f_1/n, the share of draws whose category appeared once, estimates the missing mass with small bias. Ecologists took the estimator as the basis of a sampling theory for species. Chao (1984) bounded the number of unseen species from f_1 and f_2, and Chao and Jost (2012) wrote the expected coverage of a sample enlarged by m draws as a closed form in n, f_1 and f_2, so that a survey of a habitat can be stopped at a target coverage. Efron and Thisted (1976) asked how many words Shakespeare knew with the same machinery, and Orlitsky et al. (2016) gave the optimal prediction of the number of unseen species for a sample up to about $n \log n$ times larger than the one in hand.

The estimators have reached text and machine learning a few times. Hajič and Moss (2025) apply the Chao bound to a catalogue of chant to say how far a collection is from complete. Tran et al. (2017) fit species models to the answers to an open ended survey question to predict when new answers stop appearing. Chung et al. (2018) count the unknown examples of a training set with the same estimator, a species being a unique example, and Pal et al. (2026) estimate the mass of deployment states a classifier has never seen. In all four a species is a discrete label and two items either match or do not.

2.2 The missing mass in a metric space

Maurer (2022) removed the discreteness. In a metric space with distance $d(\cdot, \cdot)$, the missing mass of a sample $X_1, \ldots, X_n$ at scale r is the probability that a new draw lies farther than r from every X_i. Maurer calls it the conditional missing mass because it is conditional on the sample drawn. His estimator is

$$G(X, r) = \frac{1}{n} \left| \{k : d(X_k, X_i) > r \text{ for all } i \neq k\} \right|, \tag{1}$$

the share of sample points with no other sample point within r. In the discrete case, with $d(x, y) = 1$ for $x \neq y$ and $r < 1$, it is f_1/n. Maurer proved that G exceeds the conditional missing mass by at most $1/n$ in expectation and that the variance of G minus the conditional missing mass is at most $3/n$ (his Theorem 2.1). The paper is theory. It sketches anomaly detection and coding as applications, and it contains no data, no embeddings and no statement about how G falls with n. The nearest ecological construction is the attribute diversity of Chao et al. (2019), in which two species closer than a threshold τ count as partly the same and diversity is reported as a function of τ.

We treat r the same way, as a resolution the user chooses.

2.3 Diversity of a sample in embedding space

Several measures describe how spread out a set of embedded points is. The Vendi score is the exponential of the entropy of the eigenvalues of the sample's cosine kernel matrix, an effective number of distinct items (Friedman and Dieng, 2023), and its cousins extend it to the orders of a Hill number (Pasarkar and Dieng, 2024). The precision and recall of Kynkäänniemi et al. (2019) count generated points inside nearest neighbour balls around a reference set, the density and coverage metrics of Naeem et al. (2020) make those counts robust to the reference set's own spread, and MAUVE compares two distributions on a quantized embedding space (Pillutla et al., 2021). Each describes the sample in hand or compares two samples. None asks what a larger sample would cover, and none has a parameter in the role of a population variance.

2.4 Saturation in practice

Where a sample size for text is set at all, it is set by a stopping rule. Qualitative researchers stop interviewing when new interviews add no new code. Guest et al. (2006) found that point at twelve interviews, and a review of 23 empirical tests by Hennink and Kaiser (2022) puts it at 9 to 17. De Paoli and Mathis (2024) carry the rule to coding by a language model and track the slope of the count of unique codes. TopicGPT reports that its topic count levels off after about 600 documents and stops when 200 documents add no topic (Pham et al., 2024). Maeda et al. (2025) fit an LDA model with eight topics to 522 open ended survey responses and find that a sample of 100 responses recovers over 60% of the topics found on the full corpus and 250 recover about 70%. These are observations of a coverage curve. The present paper gives the curve a parameter, the intrinsic dimension d, and an estimator.

3 Method

3.1 Setting and estimand

A corpus is a finite set of N documents $x_1, \ldots, x_N$. Each is embedded by a fixed encoder and normalised, and the distance between two documents is the cosine distance of their embeddings. A sample S is a set of n documents drawn without replacement. The user chooses a resolution r: two documents within r of each other count as the same content for the purpose at hand. The quantity to estimate, for the sample actually drawn, is the missing mass at resolution r,

$$M(S, r) = \frac{1}{N} \left| \{ j : d(x_j, x_i) > r \text{ for all } i \in S \} \right|, \tag{2}$$

the share of corpus documents with no sampled document within r. Coverage is $1 - M$. The estimator is Maurer's G of Equation 1, computed on the sample alone from each sampled document's distance to its nearest sampled neighbour. We write $G(n, r)$ for its value on a sample of size n.

The estimand is the finite corpus in hand. Sampling exists because the corpus is too large to annotate or to place in a model's context, and the full corpus is available to check the estimate. This is the setting of the experiments below, and it is what makes a finite population correction necessary.

3.2 The collapse law

Suppose the embedded documents lie near a manifold of intrinsic dimension d with a density that varies smoothly. The mass of a ball of radius r around a document is then proportional to r^d, with a constant that depends on the local density, and the probability that none of n sampled documents falls in the ball is about $\exp(-c\, n\, r^d)$ for that constant c. Averaged over documents, the isolation rate G should then depend on n and r only through the product nr^d. A density that varies across the manifold turns the single exponential into an average of exponentials with different rates. That average falls more slowly than any one exponential, and a stretched exponential, $\exp(-(c\, n\, r^d)^b)$ with $b < 1$, describes it well. We therefore fit

$$\log\big(-\log G(n, r)\big) = a + b \log\big(n\, r^d\big), \tag{3}$$

with d estimated beforehand and a and b fitted by least squares. The cells are an (n, r) grid of eleven sample sizes from 100 to 7,000 and twelve radii spaced geometrically from the 5th to the 99th percentile of a pilot's nearest neighbour distances, with one nested sample per n, and only cells with $0.02 < G < 0.98$ enter the fit. A value $b = 1$ is the pure exponential and $b < 1$ is the stretched form.

The dimension d comes from the TwoNN estimator of Facco et al. (2017), which uses only the ratio of each point's second to first nearest neighbour distance, on a pilot of 2,000 documents. As a check that owes nothing to TwoNN, we also regress

$\log(-\log G)$ on $\log n$ and $\log r$ separately. If the law holds, the coefficient on $\log r$ is $b\,d$ and the coefficient on $\log n$ is b, so their ratio estimates d.

Solving Equation 3 for n gives the sample size at which the isolation rate at resolution r reaches a target ε,

$$n^*(\varepsilon, r) = \left(\frac{\log(1/\varepsilon)}{e^a}\right)^{1/b} r^{-d}. \qquad (4)$$

This is the corpus analogue of $n = z^2\sigma^2/e^2$. The target ε plays the part of the tolerated error, the resolution r is set by the question, and d is the property of the corpus that the formula needs, in the role of the variance. Each halving of r multiplies n^* by 2^d.

3.3 Finite population correction

G estimates the missing mass of the distribution that generated the corpus. When the estimand is the corpus itself, the n sampled documents are covered at distance zero, and each of the $N - n$ unsampled documents is isolated from the sample with probability close to G. The corrected estimator is therefore

$$\hat{M}(n, r) = G(n, r)\,\big(1 - n/N\big), \qquad (5)$$

with $1 - n/N$ the finite population correction of survey sampling. The correction is negligible when n is a small share of N and halves the estimate at $n = N/2$.

3.4 A diagnostic for corpora with more than one scale

Equation 3 assumes one dimension for the whole corpus. The R^2 of the fit, and the R^2 on cells with $n > 1{,}000$ when the fit uses only cells with $n \leq 1{,}000$, say whether the assumption holds. When it fails, two probes locate the cause: the share of documents with a near duplicate (a nearest neighbour within cosine distance 0.02), and the spread of the TwoNN dimension across twenty k-means clusters of the corpus.

3.5 Theme coverage as a question about a partition

A different question is how many documents are needed to see every theme. Once themes are a partition of the corpus with masses $m_1, \ldots, m_K$, the expected mass of themes present in a sample of n draws with replacement is

$$C_K(n) = \sum_{k=1}^{K} m_k\,\big(1 - (1 - m_k)^n\big), \qquad (6)$$

the Good-Turing expectation, and it depends on the corpus only through the masses. Our samples are drawn without replacement, and at $n \leq 400$ from corpora of at least 7,600 documents the difference is below 0.003. We test the equation with inventories of $K = 30$ themes per corpus, against the exact label coverage of each sample and against the themes a language model reports from the sample. Section 5.5 shows that this curve and the missing mass at a fixed r behave differently across corpora, which is why the two questions need two tools.

4 Data

Table 1 lists the corpora. Two synthetic families have a known answer. Gaussian manifolds place 20,000 points on a random d-dimensional subspace of $\mathbb{R}^{64}$ for $d \in \{2, 4, 8\}$, so the true intrinsic dimension is known and the collapse law can be checked against it. Zipf topic mixtures draw 20,000 documents from 50 to 500 topic centres with Zipf weights, isotropic Gaussian noise at scale 0.6 or 1.0, and normalisation to the unit sphere, so the topic coverage of any sample is known exactly.

Ten real corpora come from four sources. Three forum corpora come from the evaluation of a corpus summarizer (Morris et al., 2023): AskDocs (medical questions from Reddit), Trump (posts about the 2020 election), and TIFU (stories told in the first person), 10,000 documents each after cleaning and deduplication, embedded with GTR-T5-base (Ni et al., 2022) at three random seeds. Five corpora are keyword slices of a 2020 export of tweets about the United States Census: random, citizens, trump, covid and illegal, 7,610 to 30,136 tweets after URL and mention removal and deduplication. Two corpora are uniform samples of 50,000 English posts each from Truth Social and from Bluesky between May and December 2024, drawn from full platform collections. The Truth Social collection is conditioned on the 2024 candidates and the Bluesky collection is a broad political pool. The census, Truth Social and Bluesky corpora are embedded with Qwen3-Embedding-8B (Zhang et al., 2025), 4,096 dimensions, 256 tokens, normalised, and the three forum corpora are embedded with the same model a second time for the theme study. Cosine distance is used throughout.

For the theme study each of the eight forum and census corpora received an inventory of 30 themes: k-means on the full Qwen3 embeddings, each cluster named and defined by Gemma 4 (Gemma

Corpus	N	Encoder	d
Manifold d=2	20,000	none	2.0
Manifold d=4	20,000	none	4.2
Manifold d=8	20,000	none	7.6
Zipf, 200 topics	20,000	none	11.0
AskDocs	10,000	GTR	13.4
Trump	10,000	GTR	9.3
TIFU	10,000	GTR	17.7
Census tweets, 5 slices	7,610 to 30,136	Qwen3	7.9 to 13.0
Truth Social	50,000	Qwen3	12.3
Bluesky	50,000	Qwen3	15.8

Table 1: Corpora. d is the TwoNN intrinsic dimension of a 2,000-document pilot, averaged over seeds where several exist.

Team, 2026) from its 15 documents nearest the centroid. The same model summarized or enumerated 1,200 samples, 30 independent draws at each $n \in \{25, 50, 100, 200, 400\}$ per corpus, at temperature 0, with documents truncated to 600 characters. Prompts are in Appendix E.

5 Results

5.1 The estimator tracks true topic coverage

The Zipf mixtures give every document a topic, so the true topic mass covered by a sample is known. Over four settings (50 to 500 topics, two noise levels), three seeds and eleven sample sizes from 50 to 5,000, coverage $1 - G$ at a radius of half the median pairwise distance of a pilot tracks the true coverage with Spearman $\rho = 0.968$ and a mean absolute error of 0.029 (132 sample points). Good-Turing on the true labels, which G does not see, reaches $\rho = 0.989$ and error 0.016. At a tenth of the median distance the agreement is gone ($\rho = 0.31$). The radius matters, and the next result gives the dependence on it.

5.2 The isolation rate collapses on nr^d

Table 2 reports the fit of Equation 3 with d fixed at its TwoNN estimate. On the three manifolds the fit explains 95% to 97% of the variance of $\log(-\log G)$, against 4% to 6% for $\log n$ alone and 64% to 73% for $\log r$ alone. On AskDocs, TIFU, the random census tweets, Truth Social and Bluesky it explains 95% to 99%. Fitted on cells with $n \leq 1{,}000$ and tested on cells with n up to 7,000, the law keeps R^2 between 0.93 and 0.99 on these five corpora, and the G it predicts at $n = 7{,}000$ has a mean absolute error of 0.044 on AskDocs, 0.029 on TIFU, 0.051 on the tweets, 0.039 on Truth Social and 0.018 on Bluesky. Across seeds 42 to 44 on the forum corpora the out of sample R^2 is 0.950 (sd 0.019) on AskDocs and 0.954 (sd 0.004) on TIFU. Figure 1 shows the surfaces against nr^d.

The fit with $\log n$ and $\log r$ as separate regressors recovers the dimension without TwoNN. The ratio of the two coefficients is 2.1, 4.4 and 7.3 on the manifolds of true dimension 2, 4 and 8, and 15.4 and 19.2 on AskDocs and TIFU against TwoNN values of 13.6 and 17.7, within 15% in every case. The exponent b is below 1 on every corpus: 0.60 to 0.75 on the Gaussian manifolds, whose density falls off from the centre, 0.45 on AskDocs and TIFU, 0.46 on Truth Social, 0.38 on Bluesky and 0.23 on the tweets.

The sample size formula was tested directly. For each corpus, Equation 3 was fitted on cells with $n \leq 1{,}000$, Equation 4 gave n^* for targets $\varepsilon \in \{0.3, 0.2, 0.1\}$ at every radius where the surface crosses ε beyond $n = 1{,}000$, and the prediction was compared with the crossing interpolated from the grid. On AskDocs, TIFU and the three manifolds, 20 of 24 predictions lie within a factor of 1.5 of the empirical sample size and all 24 within a factor of 2 (median ratios 0.93, 1.06, 1.21, 1.38 and 0.97). On Trump and the Zipf mixture, which fail the collapse, the predictions are off by factors of 3 and 80 (Appendix F).

5.3 Where the collapse fails

Trump and the Zipf mixture fail the fit (out of sample R^2 0.64 for both, and 0.56 with sd 0.09 for Trump across seeds). Two probes locate the difference. In a 5,000-document sample of Trump, 3.4% of documents have a near duplicate within cosine distance 0.02, against 0.0% in AskDocs and TIFU and below 0.1% on Truth Social and Bluesky. The TwoNN dimension of Trump's twenty k-means clusters runs from 2.0 to 16.2 (coefficient of variation 0.52), against 0.15 for AskDocs, 0.10 for TIFU, 0.16 for Truth Social and 0.13 for Bluesky. The clusters of dimension near 2 hold slogans and the clusters near 16 hold discussion. Removing the 266 documents of the full corpus that have a near duplicate leaves the out of sample R^2 at 0.74, so duplication accounts for part of the failure and the spread of local dimension is the remaining candidate. The Zipf mixture has no duplicates and its local dimension varies less (coefficient of variation 0.21), but its topic masses span two orders of magnitude, so the density around a document varies by the same factor. Both corpora have more than one

Corpus	d (TwoNN)	R^2, nr^d	R^2, n only	R^2, r only	out of sample R^2	error at n=7,000
Manifold d=2	2.01	0.970	0.060	0.635	0.952	0.032
Manifold d=4	4.21	0.954	0.044	0.666	0.924	0.056
Manifold d=8	7.60	0.971	0.054	0.731	0.939	0.031
AskDocs	13.6	0.973	0.023	0.801	0.931	0.044
TIFU	17.7	0.972	0.041	0.826	0.956	0.029
Census tweets (random)	14.5	0.946	0.033	0.863	0.939	0.051
Truth Social	12.3	0.966			0.927	0.039
Bluesky	15.8	0.992			0.991	0.018
Trump	9.2	0.811	0.000	0.731	0.636	0.064
Zipf, 200 topics	11.0	0.662	0.003	0.558	0.639	0.141

Table 2: Collapse of the isolation surface. R^2 of $\log(-\log G)$ on $\log(nr^d)$, on $\log n$ alone and on $\log r$ alone over all grid cells with $0.02 < G < 0.98$; the out of sample R^2 and the mean absolute error of the predicted G at $n = 7{,}000$ come from a fit on cells with $n \leq 1{,}000$. Seed 42; seed means for the forum corpora are in the text. The columns with a single regressor were not run for Truth Social and Bluesky.

n	M (true)	G	$G\,(1 - n/N)$
100	0.729	0.749	0.743
200	0.665	0.684	0.672
500	0.565	0.600	0.574
1,000	0.479	0.534	0.486
2,000	0.378	0.463	0.380
5,000	0.211	0.384	0.212

Table 3: Missing mass of the finite corpus, mean over four corpora and six radii at each n. N is 10,000 for the three forum corpora and 20,000 for the Zipf mixture.

scale, and one d cannot describe them. The R^2 of the collapse fit is therefore a diagnostic: a corpus that collapses has one dimension and one sample size formula, and a corpus that does not fails the formula, as the tests of n^* above show.

5.4 The finite population correction

Table 3 compares G and $G\,(1 - n/N)$ with the true missing mass M of the finite corpus over four corpora (AskDocs, Trump, TIFU and the Zipf mixture), six sample sizes from 100 to 5,000 and six radii, 144 cells. Uncorrected, G has a mean absolute error of 0.068 and a maximum of 0.42, reached at $n = 5{,}000$ of $N = 10{,}000$, where G is twice M. Corrected, the mean error is 0.015, the maximum 0.11 and the Spearman correlation 0.996. Across the three seeds on the forum corpora the corrected error is 0.010 on AskDocs, 0.013 on TIFU and 0.009 on Trump, and on Truth Social and Bluesky, where n/N is at most 0.1, it is 0.011 and 0.013. The correction holds on the corpus that fails the collapse as well.

5.5 Theme coverage is a question about a partition

With thirty themes per corpus, the share of theme mass present in a sample is the same in all eight forum and census corpora at every n: 0.58 to 0.65 at $n = 25$, 0.82 to 0.86 at $n = 50$, 0.94 to 0.98 at $n = 100$ and 0.99 to 1.00 at $n = 200$ (Figure 2a). Equation 6, evaluated on each corpus's theme masses, reproduces these values with a mean absolute error of 0.005. Over the same samples, coverage at a fixed resolution, $1 - G$ at $r = 0.34$ (the median nearest neighbour distance in 50-document samples pooled over the eight corpora), runs from 0.003 on AskDocs to 0.77 on the citizens tweets at $n = 25$ (Figure 2b). The Spearman correlation across the eight corpora between the two quantities is $+0.12$ at $n = 50$ and $+0.07$ at $n = 200$.

The language model's enumeration behaves as the partition says it should. The model answers with the identifiers of inventory themes it finds present, each with a supporting document. Within each corpus the mass of themes it reports rises with n along the label coverage curve, from 0.44 to 0.60 at $n = 25$ to 0.96 to 1.00 at $n = 400$, and the share of reported themes that the labels confirm rises from 0.73 at $n = 25$ to 0.98 at $n = 400$. Merging the eight inventories into one shared taxonomy, by clustering their centroids at four cosine thresholds (42 to 187 shared themes), gives corpora different shared coverage at the same n. Equation 6 on the shared masses predicts those differences (Spearman $+0.93$ to $+1.00$ across corpora at every n and threshold), and G at a fixed radius does not ($+0.31$, $+0.21$, $+0.38$ and -0.40 at $n = 50$ for the four thresholds). Appendix C gives the full table.

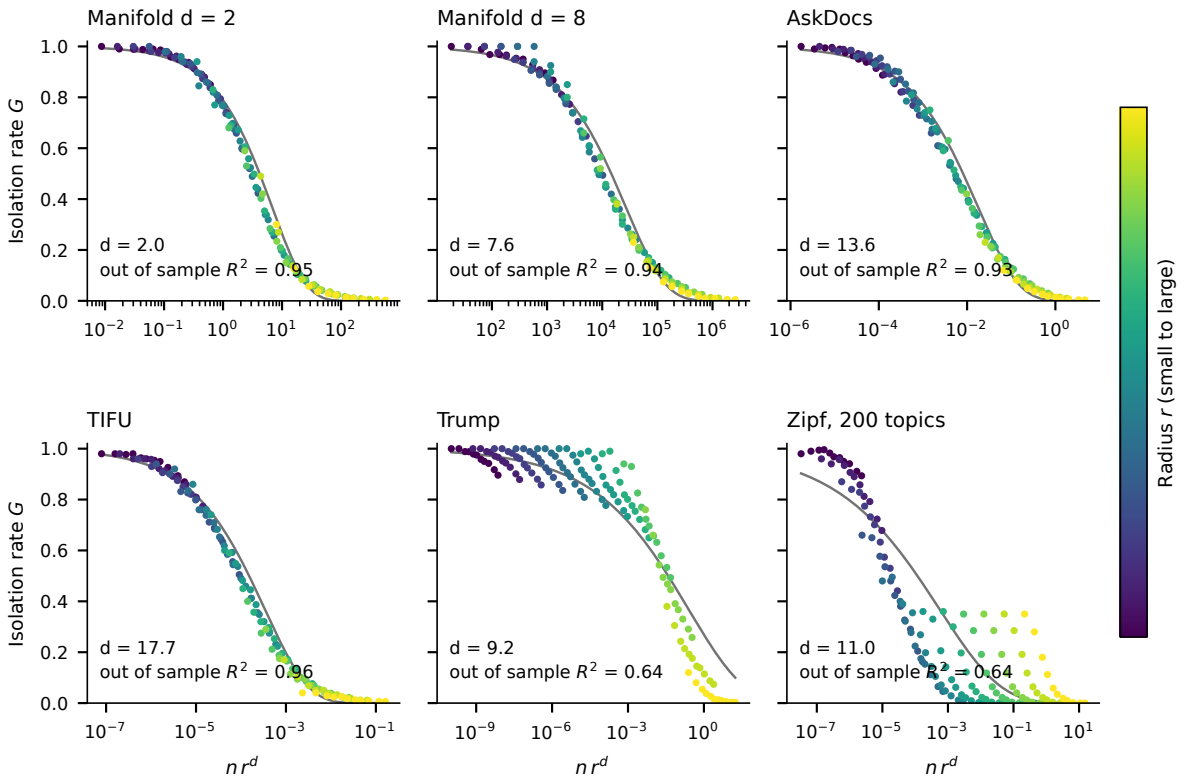


Figure 1: Isolation rate G against nr^d on a log scale, one panel per corpus, colour by radius r. Points from different radii fall on one curve where the law holds (AskDocs, TIFU, the manifolds) and separate where it fails (Trump, the Zipf mixture). The grey line is the fit of Equation 3.

The two questions are therefore different functionals of the same geometry. Coverage of a partition is set by the partition's mass vector. The missing mass at resolution r is set by the local geometry at scale r. The two coincide when the partition's cells are balls of radius r, and thematic clusters are not. A user who asks "have I seen every theme" should use Equation 6 with the theme masses, which a pilot inventory supplies. A user who asks "is every document within r of something I have read" should use G, the collapse law and Equation 4. The stability of a free summary written from a sample is a third measure, and Appendix B shows that it follows coverage within a corpus and does not order corpora.

5.6 Truth Social and Bluesky

The two platform samples, 50,000 posts each, obey the law. Truth Social has $d = 12.3$ and Bluesky $d = 15.8$. The collapse explains 97% and 99% of the variance, the out of sample R^2 is 0.93 and 0.99, and the predicted G at $n = 7{,}000$ errs by 0.039 and 0.018. Fewer than 0.1% of posts have a near duplicate and the local dimension across twenty

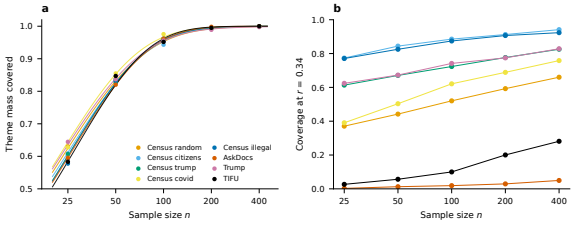


Figure 2: (a) Share of theme mass present in a sample against n, eight corpora with thirty themes each (points), and the Good-Turing expectation of Equation 6 from each corpus's theme masses (lines). (b) Coverage at the fixed resolution $r = 0.34$ for the same samples.

clusters has a coefficient of variation of 0.16 and 0.13. At the same resolution, Bluesky's higher d means that a target isolation rate needs more posts there than on Truth Social, by a factor of $r^{-3.5}$.

6 Discussion

The procedure a user follows has four steps. Choose the resolution r that matches the question: a coarse r for "are the main kinds of post represented", a fine r for "has every specific claim been seen". Embed a pilot of about 2,000 documents

and estimate d by TwoNN. Fit Equation 3 on nested subsamples of the pilot and check its R^2. Take n^* from Equation 4 and, after the sample is drawn, report $G\,(1 - n/N)$ as the share of the corpus the sample leaves uncovered. No step needs labels, and no step needs a GPU once the embeddings exist.

The survey analogy holds in two places and is instructive in a third. The finite population correction is the same factor for the same reason. The dimension d plays the role of the variance: it is the one property of the population that enters the sample size formula, and like the variance it is estimated from a pilot, with a and b fitted alongside it. The third place is the margin of error. A survey's margin bounds the error of a proportion, and a corpus sample's plain mean already does that job well, to within 0.01 for cluster proportions at $n = 1{,}000$ (Appendix D). The missing mass bounds something else, the mass of content the sample has no example of, which is the error that matters for enumeration: listing the themes, finding the rare claim, checking that a summary had the chance to mention something. A report of a corpus sample should carry both.

The theme result explains two things an applied reader will have met. A free summary written from a sample is stable across draws at a rate set by the summarizer's own granularity (Appendix B), and a theme inventory of fixed size is covered at the same rate in every corpus. Both measures are relative to the corpus, and neither can see how broad the corpus is at any absolute scale. The breadth that the sample size formula is about lives at a resolution, and only a measure at that resolution can see it.

The collapse diagnostic has a use beyond the formula. A corpus whose isolation surface does not collapse holds material at more than one scale, slogans and discussion in the Trump corpus, and a single sample size will oversample one and undersample the other. The probe of local dimension across clusters locates the parts, and a stratified sample with a size per part is the next step, which we have not tested.

Limitations

The resolution r is a cosine distance in one encoder's space, and the same r means different things in GTR and in Qwen3 embeddings. The collapse law held under GTR on the forum corpora and under Qwen3 on the census, Truth Social and Bluesky corpora, and d moved with the encoder (13.6 under GTR against 12.6 under Qwen3 on AskDocs), so a user must fix the encoder before choosing r and must report the encoder with r.

The exponent b of the stretched exponential is fitted. We have an account of why it is below 1 but no closed form that predicts it from the density, so the formula needs a pilot large enough to fit two coefficients.

The corpora are English social media and forum text from four sources. The Truth Social collection is conditioned on candidate threads, so its d describes that collection and not the platform.

The stability of summaries written from samples, measured as the mean pairwise distance among the summaries of thirty independent samples, falls with n along the coverage curve within each of eight corpora, and across corpora at a fixed n the ordering by coverage does not predict the ordering by stability (Appendix B). A summary's stability depends on the scale at which the model abstracts, which this paper does not model.

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A Synthetic smoke test

Zipf topic mixtures with 20,000 documents in 64 dimensions, four settings (50 topics with Zipf exponent 0.5; 200 and 500 topics with exponent 1.0; 200 topics with noise 1.0), three seeds, nested samples at eleven sizes from 50 to 5,000. Coverage $1 - G$ at radius α times the median pilot pairwise distance,

against the true topic mass coverage: $\alpha = 0.5$ gives $\rho = 0.968$ and error 0.029; $\alpha = 0.4$ gives 0.797 and 0.091; $\alpha = 0.3$ gives 0.677 and 0.209; $\alpha \leq 0.2$ fails. The Chao and Jost extrapolation from a 500-document pilot to the sample size for 95% coverage lands within a factor of two of the empirical crossing in three of four settings and overpredicts on the noisy setting (1,267 against 1,000).

B Summary stability

For each of the eight forum and census corpora and each $n \in \{25, 50, 100, 200, 400\}$, thirty independent samples were summarized by Gemma 4 at temperature 0 in one paragraph of at most 150 words, and the summaries were embedded with Qwen3-Embedding-8B. Instability is the mean pairwise cosine distance among the thirty summaries. Within every corpus instability falls with n and tracks G (Spearman $+0.40$ to $+1.00$, seven of eight at $+0.70$ or above), but the slope of $\log$ instability on $\log G$ runs from $+0.17$ (citizens) to $+24$ (AskDocs). Across corpora at a fixed n, Spearman of instability with G is $+0.14$, $+0.12$, $+0.12$, $+0.07$ and -0.17 at $n = 25$ to 400, with bootstrap bands that include zero, and with d it is $+0.10$ to $+0.29$. The per sample G does not predict how far that sample's summary sits from the centroid (median $\rho = +0.05$ over forty cells). TIFU illustrates the mechanism: its G is near 1 at every shared radius and its summaries are the most stable, because the model summarizes it as stories of mishap whichever thirty stories it is given.

C Shared taxonomy

The eight inventories of thirty themes were merged by average linkage clustering of their centroids at cosine distance 0.20, 0.15, 0.10 and 0.05, giving 42, 69, 112 and 187 shared themes, and every document was assigned to its nearest shared centroid. Table 4 gives, for each merge and each n, the Spearman correlation across corpora of the shared theme coverage of the 1,200 samples with $1 - G$ at $r = 0.30$ and with Equation 6 evaluated on the shared masses.

Merge (themes)	n=25	50	100	200	400
with $1 - G(n, 0.30)$					
0.20 (42)	+0.26	+0.31	+0.26	+0.29	+0.38
0.15 (69)	+0.24	+0.21	−0.33		
0.10 (112)	+0.88	+0.38	−0.07	−0.24	−0.38
0.05 (187)	+0.26	−0.40	−0.40	−0.40	−0.40
with Equation 6 on the shared masses					
0.20 (42)	+0.98	+1.00	+1.00	+0.98	+0.93
0.15 (69)	+0.93	+0.98	+0.88		
0.10 (112)	+0.93	+0.90	+0.90	+0.98	+0.81
0.05 (187)	+0.98	+0.95	+0.98	+1.00	+1.00

Table 4: Cross-corpus Spearman correlation of shared theme coverage with the two predictors, eight corpora. Blank cells were not computed.

D A bound for functions that are stable at scale r

For a document function f with range 1 that takes the same value on documents within r of each other, the estimate of the corpus mean of f formed by giving each corpus document the value of its nearest sampled document errs by at most $v(r) + M(r)$, where $v(r)$ is the share of covered documents whose nearest sampled document disagrees with them on f. On a family of fifty cluster indicator functions over four corpora, six sample sizes and six radii, the bound held in 144 of 144 cells and was loose for proportions (median 22 times the realised error), where the plain sample mean already errs by 0.003 to 0.07. The bound is tight by construction for presence indicators, where the error equals the missing mass.

E Prompts

Summary. "Below are $\{n\}$ documents sampled from a larger collection. Write a summary of the collection in one paragraph of at most 150 words. Cover the main themes in proportion to how often they appear, name notable minority themes briefly, and do not mention individual document numbers."

Theme naming. "The $\{m\}$ documents below were grouped together because they are about the same theme. Give the theme a short name (at most 8 words) and a one-sentence definition that would let a reader decide whether a new document belongs to it. Answer as JSON with keys name and definition."

Enumeration. "Below are $\{n\}$ documents sampled from a larger collection, followed by a list of $\{K\}$ themes found in the full collection. Decide which themes are present among these $\{n\}$ documents. A theme is present only if you can point to one document that is primarily about it; in a small sample most themes will be absent, and similar-sounding themes are distinct, so pick the single best theme for each document. Answer with one line per present theme in the form theme_id:

document_number."

F Direct test of the sample size formula

Equation 3 was fitted on cells with $n \leq 1{,}000$; Equation 4 then gave n^* for $\varepsilon \in \{0.3, 0.2, 0.1\}$ at every radius where the surface crosses ε at some $n > 1{,}000$ within the grid, and the crossing was interpolated on the log scale between grid points. Table 5 reports the ratio of predicted to empirical sample size.

Corpus	cases	median ratio	within 1.5	within 2
Manifold d=2	9	1.21	9	9
Manifold d=4	9	1.38	6	9
Manifold d=8	8	0.97	6	8
AskDocs	6	0.93	5	6
TIFU	6	1.06	6	6
Census tweets (random)	3	2.13	0	1
Trump	2	3.00	0	1
Zipf, 200 topics	5	82	0	0

Table 5: Predicted over empirical sample size for a target isolation rate, by corpus. The tweet corpus has 7,860 documents, so few crossings fall beyond $n = 1{,}000$.